

|                                          | Reviewer 1                                                                                                    | Reviewer 2(SRout)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Reviewer 3 (LVZ)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Reviewer 4 (Luqman)                                                                                                                                                                                                                                                                                                                                                                                                                            | Reviewer 5                                                                                                                             | Reviewer 6 | Average |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------|---------|
| Comments                                 | - Vanilla low-speed SAR ADC<br>- Realistic scope definition<br>- Well-defined execution and verification plan | - 8-bit SAR-ADC is a very good open-source IP when silicon verified.<br>- Binary weighted DAC should be good enough for 8-bit<br>- 200 run monte carlo is good enough as long you see normal distro<br>- Going from 15mV to 2mV offset will require you increasing the area by $(15/2)^2$ roughly 50X assuming your Vt dominated by the input pair. That's brute force. You can use source degeneration to reduce the area req.<br>- Good documentation along with a | - Classical topology for the SAR ADC.<br>- Comparator offset requirements depends on your reference and resolution. In your proposal that point is not clear.<br>- Also, not clear the target sampling frequency. This might affect the sizes for your capacitors.<br>- It is good that you have simulated the comparator. However, seems to have a systematic offset around 15mV.<br>- To solve systematic offset you need to size properly the next stage (inverters) after the | - Good idea to define measurable design gates early, it will help track progress throughout the project<br>- Consider validating the comparator and CDAC independently before integrating the complete SAR loop<br>- Make sure capacitor mismatch and comparator offset are included in the early Monte Carlo simulations<br>- If layout becomes the critical path, focus on completing a robust 8-bit converter before optimizing performance |                                                                                                                                        |            |         |
| Novelty score 1...4 (worst...best)       | 1                                                                                                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                        |            | 1.75    |
| Documentation score 1...4 (worst...best) | 3                                                                                                             | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                        |            | 3.00    |
| Feasibility score 1...4 (worst...best)   | 4                                                                                                             | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3.5                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                        |            | 3.63    |
| A13                                      |                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            | 2.79    |
| <a href="#">Main issue#18</a>            |                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |
| Team                                     | Project                                                                                                       | Blocks to design                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Questions/Suggestions                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |
| A13 Hungry Chippos                       | 8-bit SAR ADC                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |
| Schematic Review                         | Score (0-3)                                                                                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                                                                                                                                      |            |         |
| 1. Design Objectives & Specifications    | 1                                                                                                             | Objectives or specifications are not defined.<br>Track D: AI usage not defined.                                                                                                                                                                                                                                                                                                                                                                                      | Objectives are identified but incomplete or ambiguous.<br>Track D: AI usage ambiguous.                                                                                                                                                                                                                                                                                                                                                                                            | Objectives and specifications are mostly defined with minor gaps.<br>Track D: AI usage mostly defined.                                                                                                                                                                                                                                                                                                                                         | Clear, measurable, and well-justified design objectives and specifications.<br>Track D: AI usage well defined.                         |            |         |
| 2. Architecture & Schematic Progress     | 2                                                                                                             | Architecture not established or major portions of the design are missing.<br>Track C: Bitstream is not defined.                                                                                                                                                                                                                                                                                                                                                      | Preliminary architecture exists but significant work remains.<br>Track C: Bitstream is somewhat defined.                                                                                                                                                                                                                                                                                                                                                                          | Architecture is largely complete with most circuit blocks implemented.<br>Track C: Bitstream is partially defined.                                                                                                                                                                                                                                                                                                                             | Architecture is complete, major blocks are integrated, and design decisions are well justified.<br>Track C: Bitstream is well defined. |            |         |
| 3. Verification & Simulation             | 3                                                                                                             | Little or no verification has been performed.                                                                                                                                                                                                                                                                                                                                                                                                                        | Limited simulations or verification available.                                                                                                                                                                                                                                                                                                                                                                                                                                    | Key functionality has been verified with reasonable simulation results.                                                                                                                                                                                                                                                                                                                                                                        | Comprehensive verification demonstrating the design meets its intended specifications.                                                 |            |         |
| 4. Technical Understanding               | 1                                                                                                             | Team is unable to explain design decisions or tradeoffs.                                                                                                                                                                                                                                                                                                                                                                                                             | Basic understanding demonstrated, but important concepts are unclear.                                                                                                                                                                                                                                                                                                                                                                                                             | Good understanding of the design and technical decisions.                                                                                                                                                                                                                                                                                                                                                                                      | Excellent understanding with thoughtful analysis of tradeoffs, assumptions, and limitations.                                           |            |         |
| 5. Project Planning & Risk Management    | 1                                                                                                             | No clear development plan or risks identified.<br>Track D: Tapeout objective is not defined.                                                                                                                                                                                                                                                                                                                                                                         | Basic plan exists but significant uncertainty remains.<br>Track D: Tapeout objective is somewhat defined.                                                                                                                                                                                                                                                                                                                                                                         | Realistic plan with identified risks and mitigation strategies.<br>Track D: Tapeout objective is mostly defined.                                                                                                                                                                                                                                                                                                                               | Well-defined roadmap with clear milestones, ownership, and proactive risk management.<br>Track D: Tapeout objective is well defined.   |            |         |
| Total                                    | 8                                                                                                             | Conditional Go                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ↓ Comments ↓                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |
| Reviewer                                 | Saroj Rout                                                                                                    | - Basic specifications are not clearly mention targeted. Resolution, ENOB, conversion rate, INL and DNL are the basic specs that needs to be targeted. Now that ou have basic simulation results, targeting a realistic specification should be the goal.                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |
| Copy and Paste this ↓                    |                                                                                                               | One part of the architecture that is misunderstood is S/H. The switch along with the DAC caps serve as the S/H and the S/H simulation should                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                        |            |         |

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| <p>Reviewer: Saroj Rout</p> <p>Total score: 8 -&gt; Conditional Go</p> <p>Review sheet: [Review sheet](https://docs.google.com/spreadsheets/d/1Mvyv1pL</p> <ul style="list-style-type: none"> <li>- Basic specifications are not clearly mention targeted. Resolution, ENOB, conve</li> <li>- One part of the architecture that is misunderstood is S/H : The switch along with</li> <li>- DAC capcitor sizing based on kt/C is not shown.</li> <li>- Bootstrap switch will unecessarily incease your design time without much benef</li> <li>- A systematic offset should be reduced below LSBs. It's mostly due to the sizing</li> <li>- Remember, the mismatch data provided by the foundry is for perfectly common</li> <li>- Although each individual block status are shown, overall project tracker like Pro</li> </ul> | <ul style="list-style-type: none"> <li>- One part of the architecture that is misunderstood is S/H : The switch along with the DAC caps serve as the S/H and the S/H simulation should have the worst case DAC configuration along with compartors input cap as the load. This will give a realistic speed of the frontend.</li> <li>- DAC capcitor sizing based on kt/C is not shown.</li> <li>- Bootstrap switch will unecessarily incease your design time without much benefit for a 8-bit DAC and it will decrease your input dynamic range by Vgs unless you use a charge pump which is a project by itself. A simple transmission gate sized such that the input dependent resistance doesn't result in more than LSB of non-linearity should be a good start. If you project is ahead of schedule, then you can think of doing a bootstrap switch.</li> <li>- A systematic offset should be reduced below LSBs. It's mostly due to the sizing of the buffers unless there is some sizing mistake in the latch. Random mismatch should be first calculated which should be dominated by the input pair. Annotating your schematic with gm/ld gives a lot of insight to the circuit.</li> <li>- Remember, the mismatch data provided by the foundry is for perfectly common-centroid layout. Your current layout will result in worse offset numbers.</li> <li>- Although each individual block status are shown, overall project tracker like Prof. Kinget's review showed is going to be very helpful to track the project.</li> </ul> |
| <p>issue#18</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |